

EXERCISE 01: Polynomial Regression

Consider a probabilistic process that assigns a label y_μ to an observation x_μ with $\mu \in 1, \dots, P$. The P data are generated by drawing samples from the equation $y_\mu = f(x_\mu) + \eta_\mu$, where $f(x_\mu)$ is some fixed (but unknown) function, and η_μ is a Gaussian, uncorrelated noise variable, such that $\langle \eta_\mu \rangle = 0$, and $\langle \eta_\mu \eta_\nu \rangle = \delta_{\mu\nu} \sigma^2$. The x_μ are also uncorrelated random variable uniform on the interval $[x_m, x_M]$. We will refer to the $f(x_\mu)$ as the function used to generate the data, and σ as the noise strength. The larger σ is the noisier the data; $\sigma = 0$ corresponds to the noiseless case. To capture the structure of the dataset consider the family of functions constituted by polynomials of order α : $f_\alpha(x; \vec{\theta}_\alpha)$, where $\vec{\theta}$ is a vector containing $\alpha + 1$ parameters.

- (1) Consider two options for the (hidden) functions used to generate data:

$$f_A(x) = 2x ,$$

and

$$f_B(x) = 2x - 10x^5 + 15x^{10} .$$

- (2) Generate $P = 10$ pairs x_μ, y_μ for each case, with $\sigma = 0$ and x_μ in the range $[0 : 1]$.
- (3) Fit the data with $f_1(x; \vec{\theta}_1) = \theta_0 + \theta_1 x$, $f_3(x; \vec{\theta}_3) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$, and $f_{10}(x; \vec{\theta}_{10}) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4 + \theta_5 x^5 + \theta_6 x^6 + \theta_7 x^7 + \theta_8 x^8 + \theta_9 x^9 + \theta_{10} x^{10}$.
- (4) Generate other $P_{\text{test}} = 20$ pairs (x_μ, y_μ) for each case, with the same σ used to generate the fitted data and x_μ in the range $[0 : 1.25]$.
- (5) Check how the fits describe data in the test set.
- (6) Comment on the results.
- (7) Repeat the procedure for $P = 100$, $\sigma = 1$, $P_{\text{test}} = 20$.
- (8) How is the noise affecting the results?
- (9) Repeat the procedure again for $P = 10000$, $\sigma = 1$, $P_{\text{test}} = 100$.
- (10) What changes when the data set is larger?